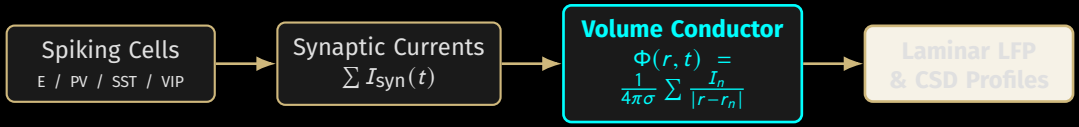
JaxFNE: Tensor-Field Neural Equations

Oscillations, Local Field Potentials, and Spectrolaminar Dynamics

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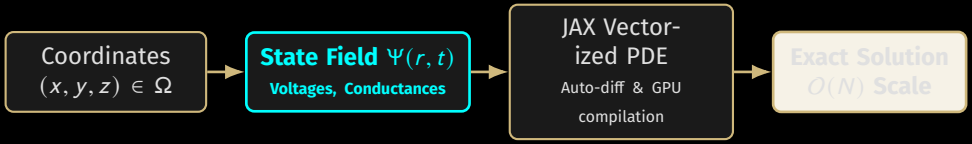
October 29, 2026

Biophysical Bridge: From Synaptic Currents to Extracellular LFP



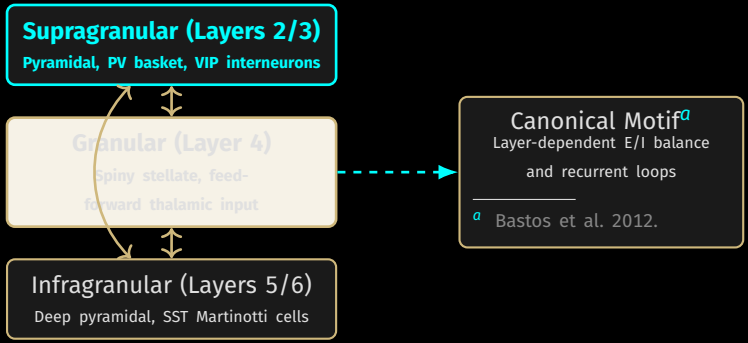
Local field potentials arise predominantly from synchronized transmembrane synaptic currents.

JaxFNE Formulation: Continuous Field Dynamics in JAX



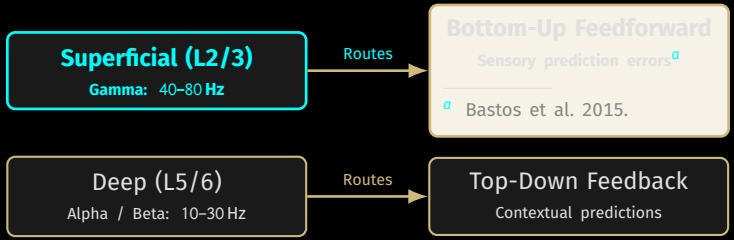
Tensor fields express continuous laminar depths without discrete connectivity bottlenecks.

Canonical Microcircuit: Layer-Specific Cortical Compartments



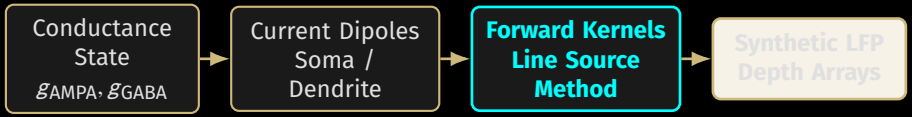
Distinct laminar compartments enforce strict division of computational labor.

Spectrolaminar Segregation: Distinct Frequency Channels



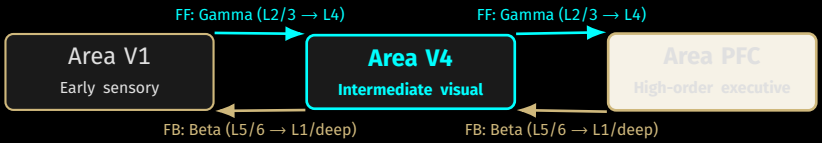
Electrophysiological power profiles mirror structural cortical hierarchy across primates.

Forward LFP Synthesis: Multi-Compartment Dipoles



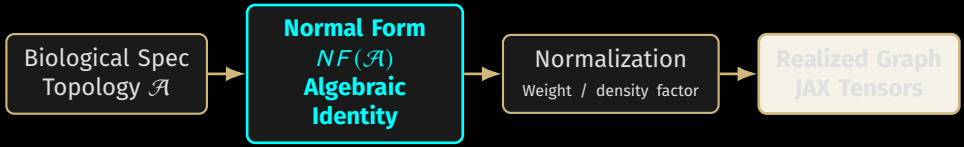
Direct resolution of extracellular field signatures from simulated neural state fields.

Cortical Hierarchy: Laminar Feedforward and Feedback Interplay



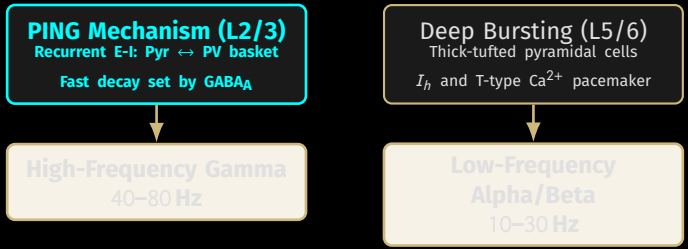
Opposing frequency streams coordinate distributed cortical processing.

TFNE Specification: Topology, Normal Form, and Compilation



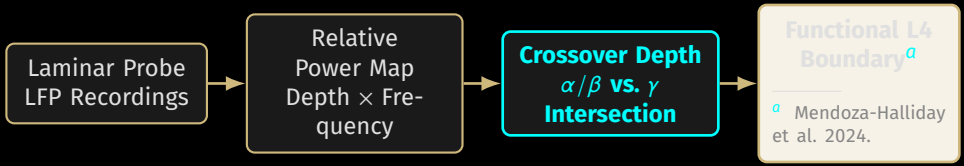
Strict separation: structural biology introduces definitions, while compilation receipts track realization.

Rhythm Genesis: PING vs. Deep Intrinsic Pacemakers



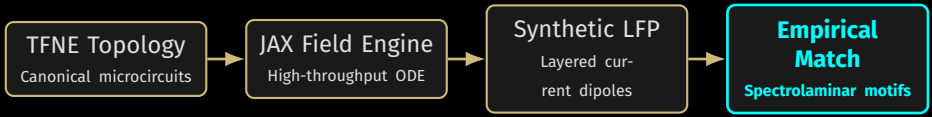
Different biophysical cellular mechanisms drive superficial vs. deep frequency bands.

Spectrolaminar Alignment: The FLIP Coordinate Mapping



Provides an objective, electrophysiological coordinate axis for cross-area cortical comparison.

Summary: Mechanistic Synthesis in JaxFNE



Computational Strength

Continuous field equations with scalable JAX hardware acceleration.

Empirical Fidelity

Reproduces laminar oscillatory and spectrolaminar biomarkers directly.